Channel Knowledge Maps for FR3 UAV Base Stations: Dataset and ML-Based Modeling

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***Abstract*—This paper introduces a dense FR3 channel knowledge map (CKM) dataset for unmanned aerial vehicle base stations. It contains 16180 ray-traced layout–height combinations from 3236 urban locations in 66 cities at 7.125 GHz. Every 513×513 realization stores building geometry, geometric visibility, attenuation, delay spread, angular spread, and a variable transmitter height. As a focused first benchmark, we propose an interpretable height-aware attenuation model. Its line-of-sight branch combines free-space loss with a coherent image-source two-ray term and compact height/range calibration. Its non-line-of-sight branch combines a COST231-shaped basis with local morphology and topology-specific linear calibration. Under a strict city holdout, the frozen model reaches 1.9287 dB overall root mean square error, with 1.7370 dB in LoS and 3.5351 dB in NLoS, on 2590 maps from 14 unseen cities. Its median CPU time is 0.0602 s per map. The result provides a transparent attenuation baseline and a physical anchor for later distribution-aware refinement of all three channel quantities.**

***Index Terms*—channel attenuation, channel knowledge map, delay spread, path loss, radio map, unmanned aerial vehicle.**

I. Introduction

Uncrewed aerial vehicles (UAVs) carrying base stations can support future 6G networks through mobility and an increased probability of line of sight (LoS). Their deployment nevertheless requires fast, location-dependent channel information for placement and resource management. Channel knowledge maps (CKMs) address this need by indexing channel information by transmitter or receiver location [1], [2].

Dense CKMs can be measured or simulated. Measurements are costly to collect at scale, whereas deterministic ray tracing (RT) provides spatially complete labels at substantial computational cost. Learned surrogates such as RadioUNet and PMNet therefore use convolutional encoder–decoders to predict dense maps from city geometry and transmitter information [3], [4]. Existing datasets, however, predominantly use fixed terrestrial transmitters and omit spatially varying building height. That combination is insufficient for elevated transmitters, whose visibility and propagation change strongly with altitude. It is especially relevant in FR3, where the 7–24 GHz range is being studied as a coverage/capacity bridge for 6G [5].

Table I compares released learning inputs rather than the provenance of the source geometry. Here, 3236 location groups paired with five transmitter heights produce 16180 realizations. "Building-height raster" means that every building pixel retains a height, rather than only a binary footprint; it does not imply that every height was surveyed directly. We combine global products providing both footprint and height [9], [10], avoiding the sparse height tags in OpenStreetMap [11].

RadioUNet introduced image-to-image dense radio-map regression with a U-Net-like architecture that encodes building geometry and transmitter location [3]. PMNet subsequently increased convolutional capacity for the same broad task [4], while more recent work uses generative models to represent fine spatial detail and uncertainty [6]. These models are appropriate when propagation effects cannot be summarized by a low-dimensional base, but their reported errors and costs depend strongly on raster support, transmitter protocol, target scaling, and split construction. A number copied across those contracts is therefore context, not a controlled ranking.

The GLOBECOM 2025 sample reviewed for this paper confirms the breadth of the current design space: two-stage diffusion [12], a heterogeneous multitask Gaussian process [13], a large masked transformer [14], a variational U-Net [15], self-supervised encoder–decoder pretraining [16], and a distribution-aware masked autoencoder [17]. Most do not disclose enough dimensions or sampling details for an exact per-map operation count. We therefore use those works to establish the architectural trend, and reserve quantitative complexity comparisons for executable models or a complete published layer table.

This work occupies a complementary point in that design space. The model is ML-based because all lookup tables and ridge coefficients are learned from training cities, but its hypothesis class is deliberately constrained: a coherent field handles repeatable LoS rings, and linear morphology features handle NLoS structure. This separation makes the stable component auditable and cheap, and leaves a physically anchored residual for a later neural refiner. The dataset is released with the richer channel targets needed to test that refinement under one common geometry and city holdout.

Increasingly expressive neural CKM estimators are valuable when sparse measurements or unresolved effects must be inferred, but their complexity is rarely isolated from stable propagation structure. We instead ask how much of the dense field can be captured by a compact physical/statistical base before learned residual refinement. The contributions are:

1) An FR3 CKM dataset from 3236 locations in 66 cities, with five transmitter heights per location and aligned attenuation, LoS, delay spread, angular spread, building geometry, and topology labels.

TABLE I
COMPARISON OF DENSE CKM AND RADIO-MAP DATASETS.

Property	This work	UrbanRadio3D [6]	RadioMapSeer [7]	CKMImageNet [8]
Variable elevated Tx height	yes	no	no	no
Layout groups or maps	3236 groups	701	701	40000
Raster size(s)	513^2	256^2	256^2	32^2–128^2
Horizontal resolution	1 m	1 m	1 m	2 m
Channel quantities	Att. (PL), LoS, DS, AS	PL, DoA, ToA	PL	Gain, AoA/AoD, delay
Building-height raster included	yes	yes	no	no
Per-sample topology labels	3 and 6 classes	no	no	no

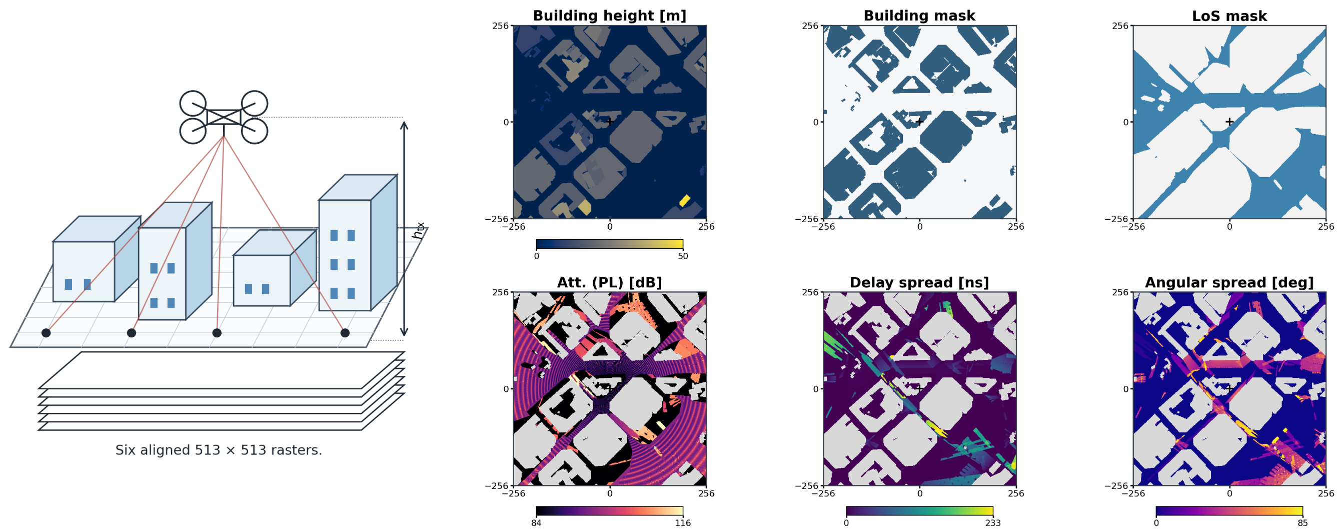

Fig. 1. One released Barcelona realization (`sample_02806`) at $h_{\text{tx}} = 69.4$ m. The left illustration shows an elevated UxNB above six aligned 513×513 rasters. The panels show building height, building mask, LoS, attenuation, delay spread, and angular spread. Crosses mark the centered transmitter; gray areas are building pixels excluded from the metrics. For this sample, the stored three- and six-class labels are `dense_highrise` and `dense_block_midrise`.

2) A height-aware attenuation model separating coherent LoS structure from topology-calibrated NLoS morphology. Every fitted quantity is frozen before evaluation on unseen cities.

The released data contain all three channel quantities; this first benchmark focuses on attenuation.

II. CKM: DEFINITION AND DATASET DESCRIPTION

For realization n, let $\mathbf{E}_n \in \mathbb{R}^{H\times W\times L}$ contain environmental layers, $\mathbf{C}_n \in \mathbb{R}^{H\times W\times F}$ contain channel quantities, and $\mathbf{m}_n \in \mathbb{R}^K$ contain scalar or categorical metadata. CKM estimation seeks a mapping

$$\widehat{\mathbf{C}}_n = \mathbf{\Phi}_{\boldsymbol{\theta}}(\mathbf{E}_n, \mathbf{m}_n), \tag{1}$$

where $\boldsymbol{\theta}$ denotes the estimator parameters. Here $H = W = 513$, $L = 2$, and $F = 4$. The environmental tensor contains

1) a building-height raster $\mathbf{H}$ in metres, with zero over ground; and
2) a binary building mask $\mathbf{B}$.

GlobalBuildingAtlas and 3D-GloBFP supply the source building geometry [9], [10]. The channel tensor, generated with MATLAB RT, stores

1) binary geometric visibility $\mathbf{V}$ (LoS $= 1$, NLoS $= 0$),
2) attenuation in dB,
3) RMS delay spread in ns, and
4) RMS angular spread in degrees.

TABLE II
DATASET SIMULATION PARAMETERS.

Parameter	Value
Carrier frequency	7.125 GHz
Antenna model	Isotropic
Map and resolution	513×513 at 1 m
Ground receiver height	1.5 m
Observed transmitter height	12 m to 478 m
Heights per location	5
Locations / realizations	3236 / 16180

Building pixels are excluded from training and evaluation. The metadata contain the centered transmitter height and stored three- and six-class morphology labels. The attenuation benchmark later recomputes an ITU-inspired three-class route from each raster; it does not use city identity or channel targets.

The 3236 locations span 66 cities and five independently sampled transmitter heights each, yielding $N = 16180$ realizations. The observed height range is approximately 12 m to 478 m; the receiver height is fixed at 1.5 m. Table II summarizes the common simulation contract, and Fig. 1 shows a released realization. The distributed CKM_Dataset_270326_labeled.h5 stores both original topology partitions and the transmitter-height bin alongside the six rasters.

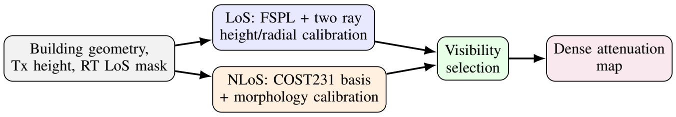

Fig. 2. Separate LoS and NLoS mechanisms are calibrated on training cities.

III. Height-Aware CKM Modeling

Fig. 2 summarizes the proposed attenuation-focused mapping. Only compact lookup tables and linear weights are learned, and all are frozen before a held-out city is evaluated.

On valid ground receivers, the stored visibility mask $\mathbf{V}$ selects

$$\mathbf{\Lambda} = \mathbf{V} \odot \mathbf{\Lambda}_{\mathrm{LoS}} + (1 - \mathbf{V}) \odot \mathbf{\Lambda}_{\mathrm{NLoS}}. \tag{2}$$

The mask is the configured MATLAB ray tracer's own binary visibility output, not a prediction of the attenuation model. For horizontal distance d_{2D}, the direct and image-source distances are $d_{3D} = [d_{2D}^2 + (h_{\mathrm{tx}} - h_{\mathrm{rx}})^2]^{1/2}$ and $d_{\mathrm{GR}} = [d_{2D}^2 + (h_{\mathrm{tx}} + h_{\mathrm{rx}})^2]^{1/2}$.

A. LoS Branch

The LoS model retains the phase difference between direct and reflected paths:

$$\Lambda_{\mathrm{FSPL}} = 20 \log_{10}(4\pi d_{3D} f/c), \tag{3}$$

$$C_{2\mathrm{ray}} = -20 \log_{10} \left| 1 + \rho(h_{\mathrm{tx}}) \frac{d_{3D}}{d_{\mathrm{GR}}} e^{-j[k(d_{\mathrm{GR}} - d_{3D}) + \phi(h_{\mathrm{tx}})]} \right|, \tag{4}$$

$$\Lambda_{\mathrm{LoS}} = \Lambda_{\mathrm{FSPL}} + C_{2\mathrm{ray}} + \beta(h_{\mathrm{tx}}) + r(d_{2D}, h_{\mathrm{tx}}). \tag{5}$$

Here $k = 2\pi f/c$; ρ and ϕ are effective calibration terms rather than material Fresnel parameters. In each 5 m training-height bin, a small grid search tests $\rho \in \{0, 0.05, \ldots, 1.5\}$ and 48 phases in $[-\pi, \pi)$. For each pair, the closed-form mean residual gives β; the pair minimizing LoS RMSE is retained. A smoothed radial table r then stores the remaining mean residual by distance. Neighboring height entries are linearly interpolated at inference. This captures the repeatable interference rings without memorizing two-dimensional building texture [18].

B. NLoS Branch

For NLoS receivers, a COST231/Hata-derived urban expression is used only as an inspectable basis function because its source domain does not cover the present frequency or heights [19]. With $d_{\mathrm{km}} = \max(d_{2D}/1000\,\mathrm{m}, 0.001)$, $f_{\mathrm{MHz}} = 7125$, and $h_{\mathrm{rx}} = 1.5\,\mathrm{m}$,

$$\begin{aligned} \widetilde{\Lambda}_C = {} & 46.3 + 33.9 \log_{10} f_{\mathrm{MHz}} - 13.82 \log_{10} h_{\mathrm{tx}} - a(h_{\mathrm{rx}}) \\ & + (44.9 - 6.55 \log_{10} h_{\mathrm{tx}}) \log_{10} d_{\mathrm{km}} + 3, \end{aligned} \tag{6}$$

$$\begin{aligned} a(h_{\mathrm{rx}}) = {} & (1.1 \log_{10} f_{\mathrm{MHz}} - 0.7) h_{\mathrm{rx}} \\ & - (1.56 \log_{10} f_{\mathrm{MHz}} - 0.8). \end{aligned} \tag{7}$$

The regression feature is $\Lambda_C = \mathrm{clip}(\widetilde{\Lambda}_C, 0, 185)$.

Let $\mathbf{G} = 1 - \mathbf{B}$ mark ground and $\mathbf{N} = (1 - \mathbf{V}) \odot \mathbf{G}$ mark NLoS ground. For a reflect-padded $k \times k$ box mean $\mathcal{A}_k$, $k \in \{15, 41\}$, define

$$\begin{gathered} \ell_d = \ln(1 + d_{2D}/1\,\mathrm{m}), \qquad \delta_k = \mathcal{A}_k[\mathbf{B}], \\ h_k = \mathcal{A}_k[\mathbf{H} \odot \mathbf{B}]/90\,\mathrm{m}, \qquad n_k = \mathcal{A}_k[\mathbf{N}], \\ \theta_n = \mathrm{clip}(\theta/90, 0, 1), \\ \sigma_{\mathrm{sh}} = 2.3197[\max(90 - \theta, 0)]^{0.2361}, \end{gathered} \tag{8}$$

where θ is the elevation angle in degrees and σ_{sh} is a deterministic angular feature, not a random shadowing realization. The implemented vector is

$$\begin{aligned} \mathbf{x} = [& \Lambda_C, \ell_d, \delta_{15}, \delta_{41}, h_{15}, h_{41}, \delta_{41}\ell_d, n_{15}, n_{41}, \\ & n_{41}\ell_d, \sigma_{\mathrm{sh}}, \theta_n, n_{41}\theta_n, 1]^\top. \end{aligned} \tag{9}$$

Routing is computed from each height raster. With building set $\mathcal{B}$, N_b four-connected footprints of area A in km^2, and one maximum height H_j per footprint, we estimate

$$\hat{\alpha} = \frac{|\mathcal{B}|}{513^2}, \quad \hat{\beta} = \frac{N_b}{A}, \quad \hat{\gamma} = \sqrt{\frac{1}{2N_b} \sum_{j=1}^{N_b} H_j^2}. \tag{10}$$

The minimum equal-weight log-ratio distance to the ITU-inspired (α, β, γ) prototypes suburban $(0.1, 750, 8)$, urban $(0.3, 500, 15)$, dense urban $(0.5, 300, 20)$, and urban high-rise $(0.5, 300, 50)$ assigns the route, with high-rise merged into dense urban [20], [21]. Height bins are low for $h_{\mathrm{tx}} \leq 58.12\,\mathrm{m}$, middle through $103.85\,\mathrm{m}$, and high above it. This gives nine topology/height regimes without a city-name lookup.

For each training map, at most 1024 NLoS pixels are selected deterministically. Non-intercept features are standardized within regime q and fitted by

$$\hat{\mathbf{w}}_q = \arg\min_{\mathbf{w}} \sum_{i \in q} (\mathbf{z}_i^\top \mathbf{w} - y_i)^2 + 10^{-2} \|\mathbf{w}_{\neg b}\|_2^2. \tag{11}$$

The deployed NLoS estimate is

$$\Lambda_{\mathrm{NLoS}} = \mathrm{clip}(\mathbf{x}^\top \hat{\mathbf{w}}_q, 20, 185). \tag{12}$$

The intercept is unpenalized. In the second expression, the coefficients have been transformed back to raw feature space.

IV. Results

A. Protocol

The city holdout contains 10840 maps from 37 training cities, 2750 from 15 validation cities, and 2590 from 14 final-test cities. Calibration uses training cities only; validation is reserved for development and ablation. Buildings are excluded before receiver statistics or metrics are computed. The test set contains 545568250 ground locations, of which 28.199% are NLoS according to the RT visibility mask. The metric further removes 122481131 ground locations without a target in the stored $[20, 185]$ dB support. Among the remaining 423087119 receivers, 7.413% are NLoS. Although the pixel-weighted score is therefore LoS dominated, 424 test maps contain more than 25% valid NLoS receivers.

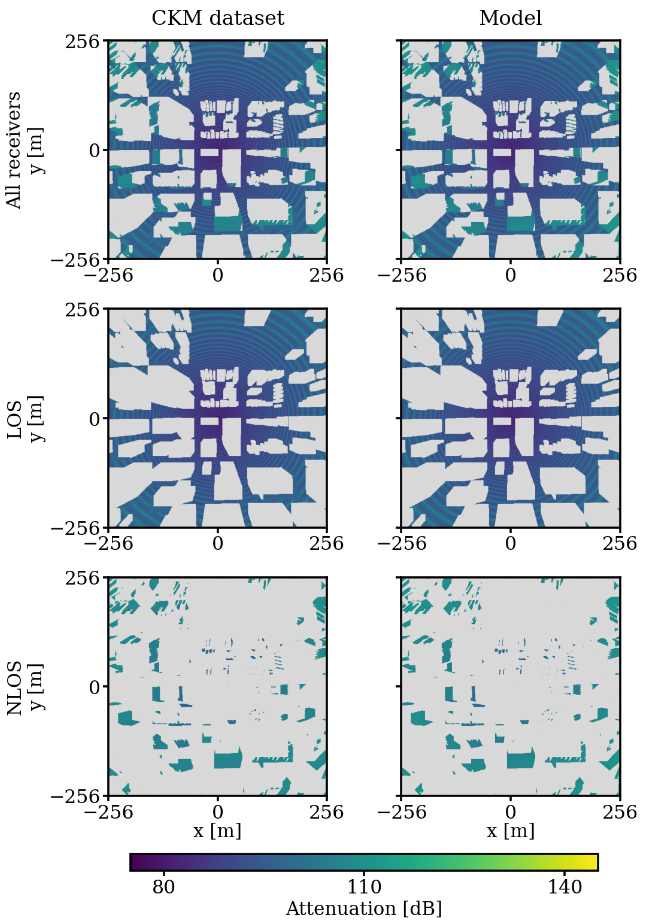

Fig. 3. Held-out Vancouver `sample_15262` ($h_{\mathrm{tx}} = 50.30$ m; urban, low-height regime). Columns compare the CKM dataset target with the model output; rows show all valid receivers, LOS, and NLOS. Gray marks buildings in the first row and receivers outside the displayed visibility subset below; building pixels never enter the metrics. The map has 119180 LOS and 25607 NLOS receivers. Full, LOS, and NLOS RMSE are 2.18, 2.10, and 2.53 dB, respectively.

For valid mask $m_i(p)$, estimate $\hat{y}_i(p)$, and target $y_i(p)$, the headline error is

$$\mathrm{RMSE} = \sqrt{\frac{\sum_i \sum_p m_i(p)[\hat{y}_i(p) - y_i(p)]^2}{\sum_i \sum_p m_i(p)}}. \quad (13)$$

Uncertainty uses 20000 percentile-bootstrap draws over held-out cities, not pixels, thereby preserving within-city spatial dependence.

B. Ablation and Accuracy

Validation ablation confirms that the coherent reflected field is essential: adding it to FSPL with height bias reduces LoS RMSE from 3.7900 to 1.7412 dB, while the radial table changes it only to 1.7388 dB. In NLoS, training-only global calibration reduces the raw COST231-basis RMSE from 25.4045 to 3.2179 dB; observable topology/height routing reaches 3.1987 dB. Thus, calibration explains the main NLoS reduction, while routing provides a smaller refinement.

TABLE III

FROZEN TEST METRICS; INTERVALS RESAMPLE 14 CITIES.

Metric	Estimate	95% interval
Overall RMSE [dB]	1.9287	[1.7984, 2.0956]
Overall MAE [dB]	1.2170	[1.1371, 1.3177]
LoS RMSE [dB]	1.7370	[1.6603, 1.8225]
NLoS RMSE [dB]	3.5351	[3.2340, 3.8457]
Macro-city RMSE [dB]	1.9829	[1.8494, 2.1292]

TABLE IV

FROZEN TEST NLOS ABLATION; EVERY ROW IS RECALIBRATED.

Variant	Routing	Terms	RMSE [dB]
Complete	topology × height	14	3.5351
Topology only	topology	14	3.5437
Height only	height	14	3.5505
Global fit	none	14	3.5568
41-pixel compact	topology × height	8	3.5618
Without 15-pixel scale	topology × height	11	3.5433
Without 41-pixel scale	topology × height	8	3.6330
Without local morphology	topology × height	5	3.6889
Topology constant	topology	1	4.5391
Global constant	none	1	4.5403

On final test, the complete model reaches 1.9287 dB overall RMSE and 1.2170 dB MAE; LoS and NLoS RMSE are 1.7370 and 3.5351 dB. A 20000-draw city bootstrap gives a 95% overall-RMSE interval of 1.7984 dB to 2.0956 dB. Per-city RMSE ranges from 1.6149 dB in Halifax to 2.5832 dB in Osaka, with an unweighted mean of 1.9829 dB.

To separate model compactness from routing, Table IV recalibrates every retained feature set on training cities before frozen test evaluation. Relative to the global 14-feature fit, topology-only routing retains 60% of the complete routing gain, whereas height-only routing retains 29%; these shares are not additive because the complete model includes their interaction. Thus, topology is the stronger routing partition on final test. The eight-feature model $[\Lambda_C, \ell_d, \delta_{41}, h_{41}, n_{41}, \sigma_{\mathrm{sh}}, \theta_n, 1]$ retains over 90% of the complete model's reduction from a single global NLoS constant. A topology-specific constant is already a useful 4.5391 dB NLoS baseline, but it is essentially tied with the global constant; topology helps primarily through its fitted slopes rather than its intercept alone.

The scale removals provide a second compactness check. Removing the 15-pixel summaries changes little, whereas removing the 41-pixel summaries causes the largest single-scale degradation. Eliminating all local morphology degrades further. Regional context is therefore useful even though the strong regime and geometry baseline makes its incremental effect look modest.

Table V uses the same low, middle, and high thresholds as the fitted model. Error decreases with height, particularly in NLoS. The topology rows also show materially different propagation regimes: dense urban NLoS is hardest, while suburban maps are most accurate. The fitted coefficient vectors change substantially across topologies, so one global set of slopes would mix distinct relationships; individual coefficients are correlated and are not interpreted as causal importance scores.

TABLE V
PIXEL-WEIGHTED TEST RMSE [dB] BY FITTED REGIME.

Regime	Maps	Overall	LoS	NLoS
Low height	846	2.1426	1.7872	3.7511
Middle height	878	1.9814	1.7849	3.5225
High height	866	1.6941	1.6595	2.6504
Suburban	1455	1.6977	1.6048	3.1628
Urban	940	2.2961	1.9967	3.6457
Dense urban	195	2.5704	2.1126	4.1275
All	2590	1.9287	1.7370	3.5351

TABLE VI
SELECTED RAW-SPACE NLOS COEFFICIENTS, MIDDLE HEIGHT.

Feature	Suburban	Urban	Dense urban
Λ_C	0.1124	0.0970	0.1334
ℓ_d	5.3587	3.3640	0.7845
δ_{41}	-29.7941	-33.5969	-34.1504
n_{41}	25.2221	3.4423	-9.4160
σ_{sh}	-10.4166	-10.7865	-10.3520
θ_n	-17.4127	-22.3150	-28.9146
Bias	132.9470	149.0716	156.2964

TABLE VII
REPORTED RMSE [dB]. ONLY THE FIRST TWO ROWS SHARE A TEST CONTRACT.

Method	Evaluation contract	All	LoS	NLoS
Proposed	CKM city holdout	1.9287	1.7370	3.5351
GMM U-Net	Same split and masks	4.3606	4.4323	3.3381
RadioUNet [3]	RadioMapSeer DPM	1.60	–	–
RadioVAE$_C$ [15]	RadioMapSeer DPM	1.42	–	–
TSRDiff [12]	BRAT-Lab, 10% samples	3.62	–	–
RO-HMTGP [13]	Measured RSRP	4.19–6.00	–	–

TABLE VIII
COMPUTATIONAL CONTEXT; REPORTED CPU ROWS USE THE SAME CPU.

Method	Grid	State	Reported MAC	CPU [s]
Proposed	513^2	66.8k+126	0.170M*	0.0602
RadioUNet [3]	256^2	13.27M	12.73G	0.1339
PMNet [4]	256^2	33.34M	52.71G	0.4042
SSL-Radio [16]	256^2	15.4–20.4M	16.9–21.9G†	–
MATLAB RT	513^2	–	–	102.289

Table VI makes the topology dependence explicit. For example, the regional blocked-ground coefficient is positive in suburban and urban maps but negative in dense urban maps. Such sign changes arise from jointly fitted, correlated regressors; they need not mean that one physical effect reverses in isolation. They do show that the fitted relationship, not only its intercept, differs by morphology.

The GMM U-Net is a controlled neural alternative: a height-conditioned mixture head and U-Net decoder, separated into six topology classes and LoS/NLoS experts, are evaluated on the same 2590 maps and receiver masks. It is the best purely neural approach found in an internal exploration of over 50 model configurations. Its lower NLoS error shows where distribution-aware learning can help, while the substantially lower LoS and overall errors of the proposed model show the value of resolving the stable coherent component explicitly. This neural result is included as a preview, not as the final hybrid: a following paper will train neural models to refine the calibrated priors for attenuation, delay spread, and angular spread.

The remaining rows are published context, not a controlled ranking. RadioUNet reports 0.020 gray-level RMSE, or 1.60 dB under its 80-dB scaling; RadioVAE$_C$ reports 0.0177, or 1.42 dB, without input measurements [3], [15]. Both use 256^2 fixed-height terrestrial RadioMapSeer data. TSRDiff uses sparse target measurements, and RO-HMTGP predicts measured RSRP across six cells rather than dense simulated attenuation. Their reported RMSE values therefore indicate scale under different tasks, not superiority over the present 513^2 variable-height city holdout.

C. Computational Comparison

Table VIII uses the same AMD Ryzen 5 5600X, six CPU threads, batch one, and GPU-disabled execution for the proposed model and neural baselines. The neural timings use official implementations, 10 warmups, and 50 forward passes; input and file I/O are excluded.

The asterisk counts the final 14-term NLoS dot product averaged over valid test NLoS pixels; feature construction and lookup operations are additional. The dagger is reconstructed from SSL-Radio's layer table under two plausible skip-concatenation interpretations, rather than measured from released code. The proposed timing includes feature construction and LoS evaluation. MATLAB R2024b takes a median 102.289 s over five Barcelona layouts with one reflection and no diffraction on the same CPU. RT also produces visibility and all three channel quantities, so the raw $\sim 1700\times$ difference is not a like-for-like speedup. It nevertheless demonstrates a significant computational gap for the attenuation prior.

D. Scope and Limitations

The reported accuracy is conditional on the supplied RT visibility mask. For fixed city geometry, transmitter position and height, receiver grid, and RT settings, that mask is deterministic and can be stored once; a repeated five-condition audit found all 742392 receiver visibility values identical. This is not a claim that an independently reconstructed mask is interchangeable with the stored one. The fitted tables also inherit the simulator, isotropic antenna, fixed carrier, flat terrain, and building products used to create the data. COST231 supplies only a calibrated basis outside its original domain, and the model does not identify the particular reflected or diffracted NLoS path. These constraints are why the result is presented as a transparent dataset baseline and refinement anchor, not as a universal propagation law.

V. CONCLUSIONS

We introduced a dense FR3 UAV CKM dataset containing attenuation, delay spread, angular spread, visibility, building geometry, transmitter height, and topology labels for 16180 realizations from 66 cities. The accompanying attenuation benchmark separates coherent LoS structure from calibrated NLoS morphology and freezes every fitted quantity before unseen-city evaluation. It reaches 1.9287 dB overall RMSE while remaining compact, auditable, and fast on a CPU. The results further show that topology is the principal NLoS calibration partition, with transmitter height refining that partition and the regional morphology scale carrying most of the useful local context.

The model is intended as a physical/statistical base rather than a rejection of deep learning. The controlled GMM U-Net result identifies NLoS as the promising refinement target, but also shows the cost of asking a network to relearn the coherent LoS field. A future journal paper will instead train distribution-aware networks to refine analogous calibrated priors for attenuation, delay spread, and angular spread. Its multi-output runtime will not be directly comparable to this attenuation-only benchmark, but the present computational gap shows that the prior-first strategy is significant and worth preserving.

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